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“

Branding is
what people
say about
you when
you are not
in the room.

”



-Jeff Bezos



```

27 // * 15.01.2023 ESP32, cleanup compiler warnings.
28
29 #include <Arduino.h>
30 #include <Wire.h>
31
32 #include <radio.h>
33
34 // all possible radio chips included.
35 #include <RDA5807M.h>
36 #include <RDA5807FP.h>
37 #include <SI4703.h>
38 #include <SI4705.h>
39 #include <SI47xx.h>
40 #include <TEA5767.h>
41
42 #include <RDSParse.h>
43
44
45 // Define some stations available at your locations here:
46 // 89.30 MHz as 8930
47
48 RADIO_FREQ preset[] = {
49   8930, // Sender:< hr3 >
50   9060, // Sender:< hr1 >
51   9310, //
52   9340, // Sender:<BAYERN 3>
53   9440, // Sender:< hr1 >
54   9530, // Sender:< YOU FM >
55   9670, // Sender:< hr2 >
56   9870, // Sender:< Dlf >
57   10020, // Sender:< planet >
58   10140, // Sender:<RADIO808>
59   10160, // Sender:< hr4 >
60   10300 // Sender:<ANTENNE >
61 };
62
63 uint16_t presetIndex = 0; ///< Start at Station with index 0
64
65
66 // ===== STA703 specific pin wiring =====

```

Fig. 5: Snippet of the source code

Testing

After proper installation, the FM signal can be heard at the preset frequency. To change the frequency, use the solder pads for SDA and SCL to reprogram the chip with a new configuration. Future enhancements (articles) could include the addition of a battery charging circuit or a small, side-mounted on/off switch for added convenience.

In next article on this subject, an SMD-based microcontroller (one of the world's tiniest MCUs) will be integrated with the FM chip to create a fully functional, ultra-compact FM radio embedded within AirPods-style earbuds. **EFY**

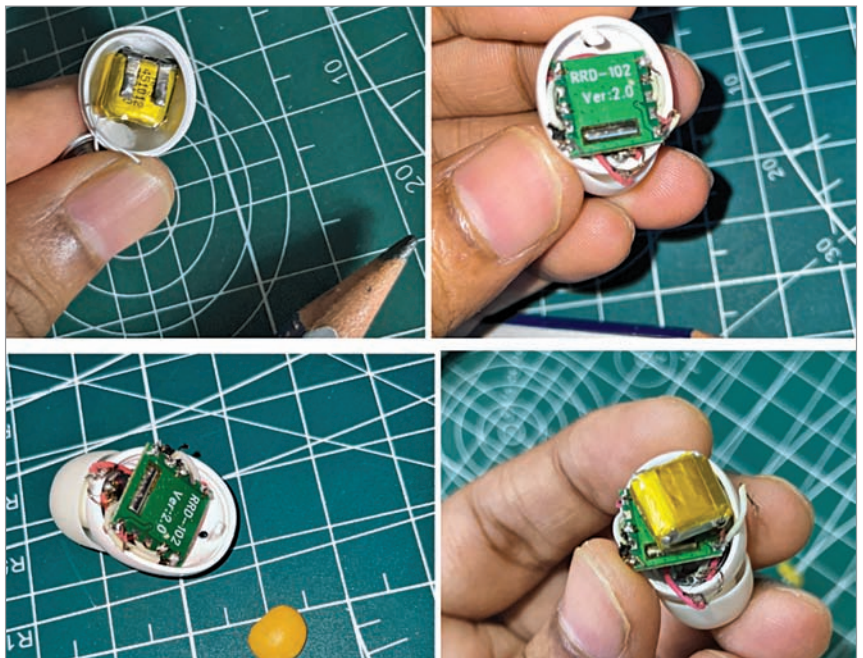


Fig. 6: Installing the battery, FM IC, and TWS earbud speaker inside the AirPods enclosure

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